UNIT IV: Environmental Pollution

Introduction

Pollution refers to any undesirable alteration in the physical, chemical, or biological characteristics of the environment air, water, or land that adversely affects living organisms and poses risks to health. As per Odum (1971), pollution is “an undesirable change in the characteristics of air, water, and land that adversely affects life and creates health hazards for all living organisms on Earth.” Southwick (1976) defined pollution as “the unfavourable alteration of the environment caused by human activities, resulting in harm to human beings and other living organisms.”

Pollution can be broadly categorized into the following two types:

1. **Natural Pollution**: This includes pollution arising from natural events such as volcanic eruptions, forest fires, soil erosion, cosmic radiation, and ultraviolet (UV) radiation.
2. **Anthropogenic (Man-made) Pollution**: This form of pollution is caused by human activities such as industrialization, urbanization, transportation, and improper waste disposal. The main categories include:
    - Air Pollution
    - Water Pollution
    - Thermal Pollution
    - Noise Pollution
    - Soil and Land Pollution
    - Radioactive Pollution
    - Marine Pollution

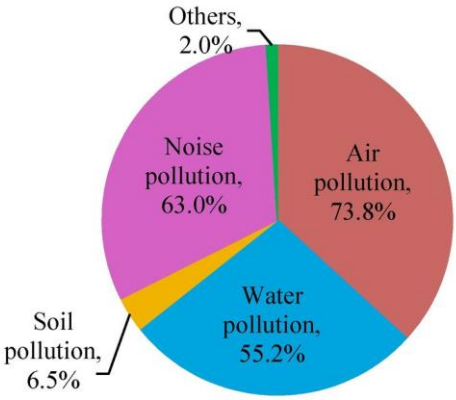

Figure 1: Various types of pollutions

Air Pollution

Air pollution refers to the contamination of the atmosphere by harmful substances that disrupt the natural composition of air, resulting in adverse effects on human health, ecosystems, and the climate. These substances, known as **pollutants**, may be present in solid, liquid, or gaseous form and originate from both natural and anthropogenic sources.

Classification of Air Pollutants

Air pollutants can be classified based on their origin and physical state:

A. Based on Origin

1. **Primary Pollutants**: Emitted directly from identifiable sources into the atmosphere. **Examples**:
   - Carbon monoxide (CO)
   - Carbon dioxide (CO_2)
   - Sulphur dioxide (SO_2)
   - Nitrogen oxides (NO, NO_2)
   - Hydrocarbons (CH_4, C_2H_6)
   - Chlorofluorocarbons (CFCs)
2. **Secondary Pollutants**: Formed in the atmosphere by chemical interactions among primary pollutants and atmospheric constituents under the influence of sunlight.

Examples:

- Ozone (O_3)
- Peroxyacetyl nitrate (PAN)
- Smog
- Acid rain (H_2SO_4, HNO_3)
- Aerosols

Figure 2: Primary Pollutants & Secondary Pollutants

B. Based on Physical State

- **Solid particles**: Dust, smoke, soot, fumes
- **Liquid droplets**: Mist, fog
- **Gaseous pollutants**: Benzene (C_6H_6), Methane (CH_4), Aldehydes(CHO), Ketones, CO, SO, NO_x

Major Air Pollutants and Their Effects

Pollutant	Chemical Formula	Source	Health/Environmental Effects
Carbon monoxide	CO	Incomplete combustion of fossil fuels	Binds to hemoglobin, reducing oxygen transport
Sulphur dioxide	SO_2	Burning of coal in power plants	Respiratory problems, acid rain
Nitrogen dioxide	NO_2	Vehicular emissions, combustion processes	Lung irritation, formation of ozone and acid rain
Ozone (Tropospheric)	O_3	Photochemical reaction	Damages lungs, crops, and rubber
Peroxyacetyl nitrate	$CH_3COOONO_2$ (PAN)	Photochemical smog	Eye irritation, respiratory issues
Chlorofluorocarbons	$CFCl_3$, CF_2Cl_2, etc.	Refrigerants, aerosol sprays	Ozone layer depletion
Suspended Particulate Matter (SPM)	$PM_{2.5}$, PM_{10}	Combustion, construction dust	Lung damage, cardiovascular diseases

Notable Secondary Pollutants

- **Ozone (O_3)**: Produced by photochemical reactions involving NO_2 and volatile organic compounds (VOCs). Though beneficial in the stratosphere, tropospheric ozone is harmful to human lungs and vegetation.
- **Acid Rain**: Formed by the reaction of SO_3 and NO_x with water vapour to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3):

  $SO_2+H_2O \rightarrow H_2SO_3$(Sulphurous acid)

  $2SO_2+O_2 \rightarrow 2SO_3$

  $SO_3+H_2O \rightarrow 2SO_2+O_2 \rightarrow 2SO_3$

  $SO_3+H_2O \rightarrow H_2SO_4$

  $2NO_2+H_2O \rightarrow HNO_2+HNO_3$

- **Smog**: A combination of smoke and fog, smog forms when pollutants such as hydrocarbons and nitrogen oxides react under sunlight. It is especially prominent in urban areas and can cause respiratory distress.

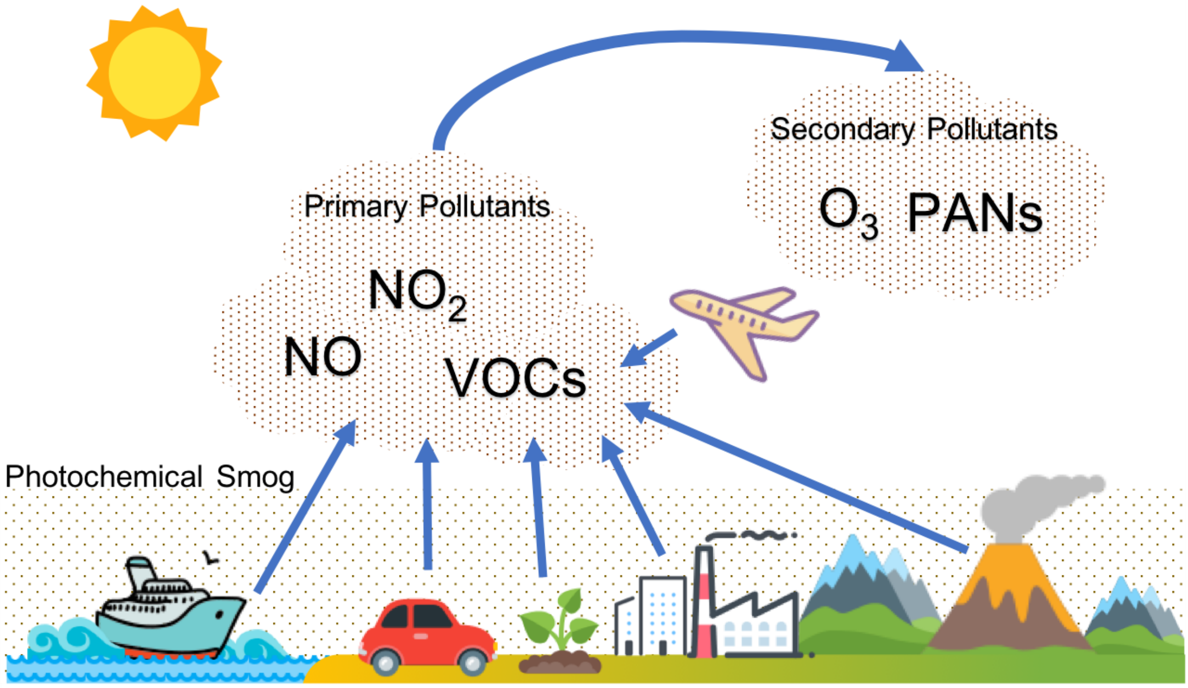

Figure 3: Smog Formation

Air Pollution Control and Prevention

1. Source Reduction and Substitution:

- Use of low-sulphur coal instead of high-sulphur variants
- Switching from fossil fuels to renewable or nuclear energy

2. Process Modification:

- Adoption of cleaner technologies and automation
- Improved combustion efficiency to minimize emissions

3. End-of-Pipe Controls:

- **Electrostatic Precipitators** – Remove particulates from flue gases
- **Scrubbers** – Clean gases using liquid spray
- **Cyclone Separators** – Use centrifugal forces to remove large particulates
- **Filters** – Capture fine particles

Water Pollution

Water pollution refers to the contamination of water bodies (rivers, lakes, oceans, and groundwater) due to the discharge of harmful substances. Pollutants may be physical, chemical, or biological, and they degrade water quality, making it unsafe for human, animal, and aquatic life.

Chemical Parameters for Water Quality Analysis

Parameter	Significance
pH	Measures the acidity/alkalinity of water. Optimal range: 6.5–8.5
Biological Oxygen Demand (BOD)	Amount of oxygen consumed by microorganisms in decomposing organic matter
Chemical Oxygen Demand (COD)	Measures total quantity of oxygen required to oxidize both organic & inorganic pollutants
Dissolved Oxygen (DO)	Indicates oxygen availability for aquatic organisms. Normal range: 5–8 mg/L

Types of Water Pollutants

1. **Pathogens**: Bacteria, viruses, and protozoa from sewage and animal waste (e.g., *E. coli*, *Vibrio cholerae*)
2. **Nutrients**: Excess nitrates (NO_3^-) and phosphates (PO_4^{3-}) lead to **eutrophication**
3. **Heavy metals**: Lead (Pb), Mercury (Hg), Arsenic (As) – toxic to aquatic life and humans
4. **Organic chemicals**: Pesticides, oil, detergents – harmful to aquatic ecosystems
5. **Thermal pollutants**: Hot water from industries lowers DO and harms aquatic organisms

Figure 4: Various types of water pollution

Eutrophication and Water Bloom

Eutrophication is the enrichment of water bodies with excess nutrients (mainly nitrogen and phosphorus), leading to dense algal blooms.

This results in:

- Oxygen depletion (hypoxia)
- Fish kills
- Loss of biodiversity
- Unpleasant odour and water taste

Water Bloom refers to the overgrowth of cyanobacteria (*blue-green algae*) on the surface of stagnant water, rendering it undrinkable.

Key Reactions

- **Decomposition of Organic Matter (BOD process)**:

  $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + Energy

  COD Testing Reactions:

  Organic matter + $K_2Cr_2O_7 + H_2SO_4 \rightarrow CO_2 + H_2O + Cr_3$

Diseases Caused by Polluted Water

Pollutant Source	Disease
Domestic sewage	Cholera, Typhoid, Dysentery
Industrial waste	Fluorosis, Minamata Disease (Hg poisoning)
Agricultural runoff	Methemoglobinemia (blue baby syndrome) due to nitrate pollution

Prevention and Control of Water Pollution

1. **Sewage treatment** before discharge
2. **Effluent treatment plants (ETPs)** for industrial waste
3. **Use of biofertilizers and biopesticides** to reduce chemical runoff
4. **Boiling and chlorination** of drinking water (30–40 mg/L chlorine)
5. **Legislation and enforcement**: Water (Prevention and Control of Pollution) Act, 1974 (India)

Noise Pollution

Definition

Noise pollution refers to the propagation of unwanted or harmful sound in the environment that disrupts the natural balance and negatively affects human health and well-being. It is commonly defined as any sound that is **unpleasant, loud, disruptive, or beyond acceptable levels**, typically measured in **decibels (dB)**.

Sources of Noise Pollution

1. Stationary Sources

- **Industries and Factories**: Mechanical operations, metal fabrication, and generators.
- **Construction Activities**: Bulldozers, cranes, jackhammers, and concrete mixers.
- **Public Address Systems**: Loudspeakers used during festivals, rallies, and public events.
- **Household Appliances**: Mixers, grinders, vacuum cleaners, and televisions.

2. Mobile Sources

- **Road Traffic**: Cars, buses, motorcycles, honking horns.
- **Railways**: Locomotive engines, track vibrations, station announcements.
- **Air Traffic**: Aircraft takeoffs and landings, particularly from jet engines.

Source Type	Example	Noise Level (dB)
Medium traffic	City roads	70–80 dB
Heavy traffic	Highways	80–90 dB
Trucks	Diesel engines	85–100 dB
Motorcycles	Inadequate silencers	90–105 dB
Jet aircraft	Takeoff	120–140 dB

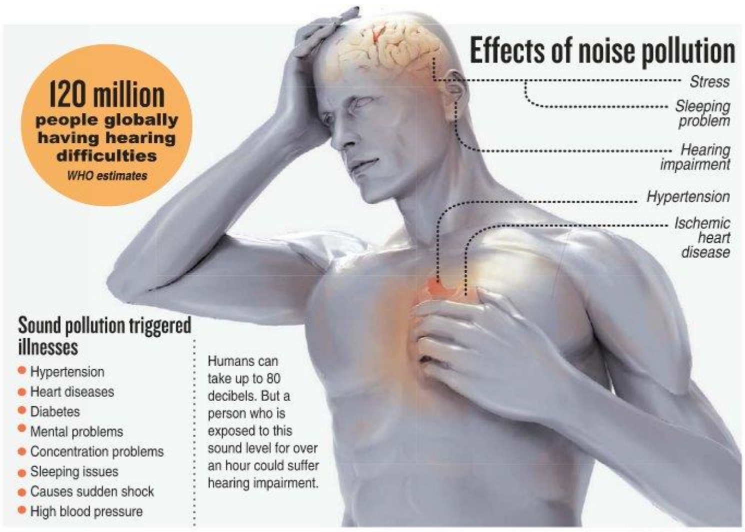

Figure 5: Effects of Noise Pollution

Effects of Noise Pollution

Affected System	Effects
Auditory System	Hearing loss, tinnitus, temporary or permanent deafness
Psychological Health	Stress, anxiety, irritability, sleep disorders
Cardiovascular System	Elevated blood pressure, irregular heartbeat
Cognitive Functions	Reduced concentration, poor academic/work performance
General Health	Headaches, fatigue, nausea, indigestion, ulcers

- **Hearing damage begins at**: > 85 dB (prolonged exposure)
- **Pain threshold**: ~120 dB

Noise Pollution in Everyday Life

Example: In an office environment, simultaneous phone rings, loud conversations, printer sounds, and external traffic noise combine to form a non-uniform, disruptive acoustic environment — classified as **noise** rather than structured sound.

Control and Prevention Measures

1. **At the Source**
   - Use of silencers in vehicles and machinery
   - Lubrication of moving machine parts to reduce vibration
   - Tightening loose parts to prevent rattling
2. **Transmission Control**
   - Acoustic insulation of buildings (double-glazed windows, soundproof walls)
   - Planting trees and establishing green belts as noise buffers
   - Erecting noise barriers along highways
3. **Regulation and Awareness**
   - Enforcement of noise-level limits (e.g., Noise Pollution (Regulation and Control) Rules, 2000 – India)
   - Prohibition of loudspeakers beyond permissible hours
   - Public education campaign

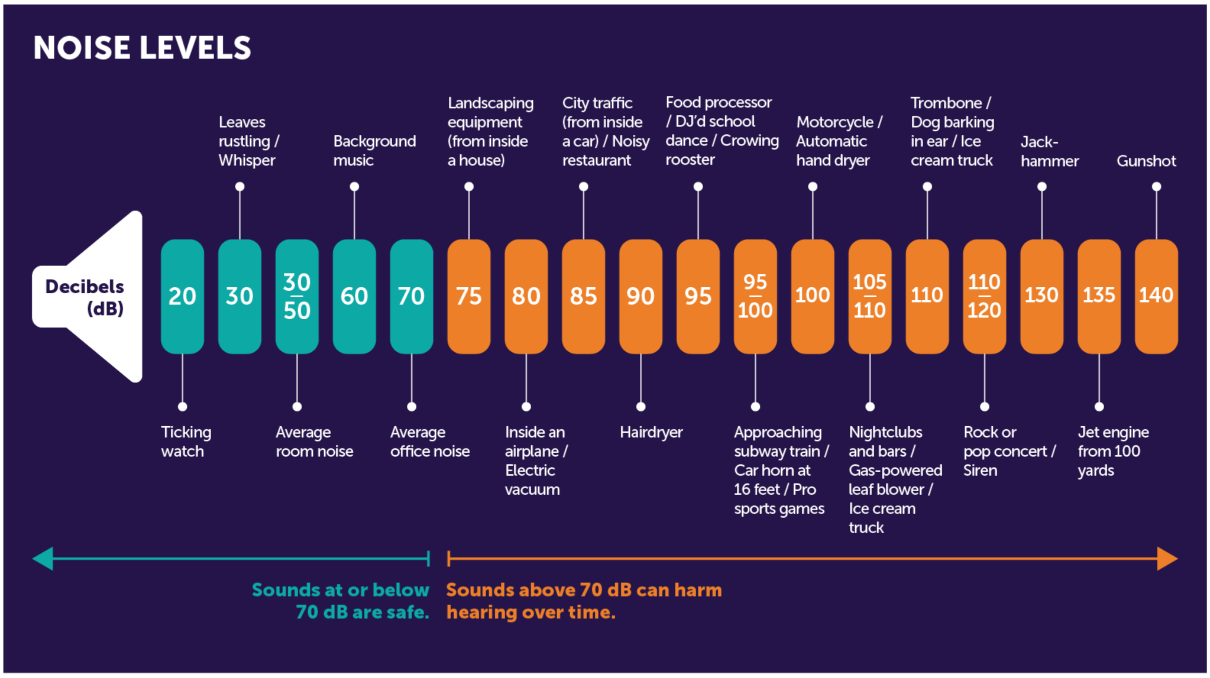

Figure 6: Noise Level

Soil Pollution

1. Introduction to Soil Pollution

Soil is one of the most vital natural resources that support life on Earth. It plays a central role in food production, water filtration, carbon storage, and biodiversity conservation. However, due to increasing human activities, soil is being subjected to pollution and degradation at an alarming rate. Soil pollution refers to the presence of toxic chemicals (pollutants or contaminants) in soil at concentrations high enough to pose a risk to human health, plant and animal life, or the environment in general. Polluted soils lose their natural fertility and productivity, leading to harmful consequences for agriculture, ecosystems, and public health.

2. Sources of Soil Pollution

Soil can be polluted by both natural and anthropogenic (human-made) sources. However, human activities are the primary contributors in modern times.

2.1 Industrial Waste

Industries discharge vast amounts of waste including heavy metals (like lead, cadmium, chromium), toxic solvents, dyes, and non-degradable chemicals into the environment.

Improper disposal and leakage into nearby landfills or water bodies result in the infiltration of these pollutants into soil.

2.2 Agricultural Inputs

Modern agriculture heavily depends on chemical fertilizers, pesticides, herbicides, and insecticides to increase crop yield. Overuse of these chemicals leads to their accumulation in soil, where they may persist for years, harming soil microbes and entering the food chain.

2.3 Municipal Waste and Sewage

Urban areas generate large quantities of solid waste and sewage. When improperly managed, these wastes are often dumped in open land or used as compost without proper treatment, contributing to biological and chemical soil pollution.

2.4 Mining Activities

Mining operations expose large amounts of minerals and metals. Waste generated from mining (tailings, slurry) often contains harmful compounds that, when dumped on land, leach into the soil and alter its chemical composition.

2.5 Oil and Fuel Spills

Oil spills during transportation or storage contaminate soil with hydrocarbons, making it unsuitable for agriculture and posing health risks through the contamination of groundwater.

2.6 E-Waste

Improper disposal of electronic waste (such as batteries, mobile phones, etc.) introduces heavy metals and toxic elements like mercury and arsenic into the soil.

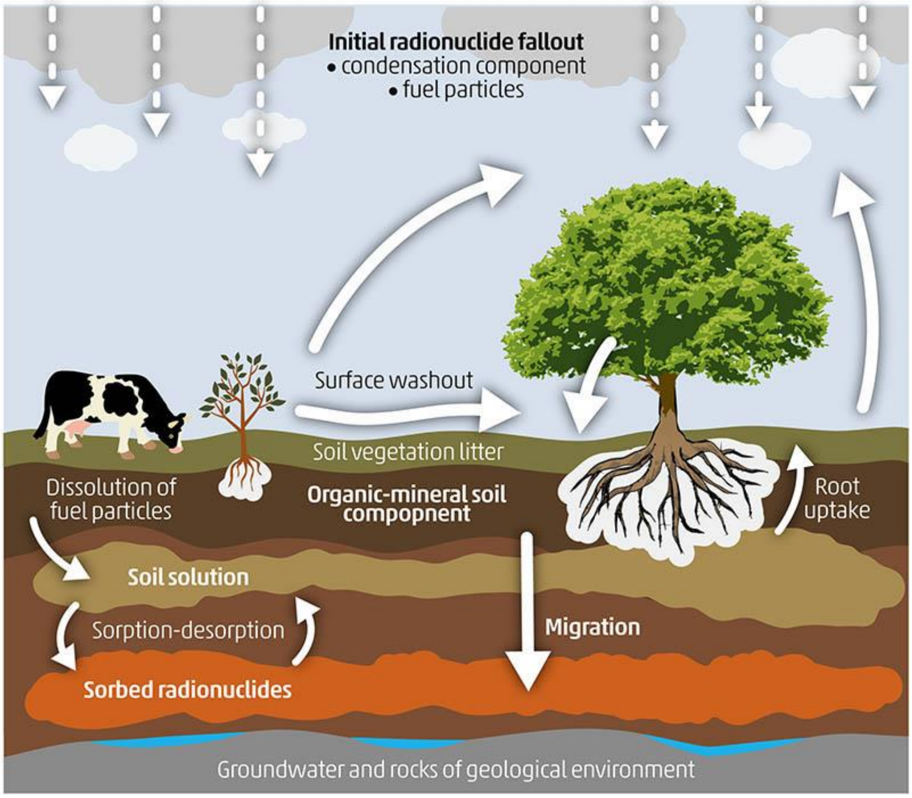

Figure 7: Sources of soil pollution

3. Types of Soil Pollution

Soil pollution can be classified based on the nature of pollutants:

3.1 Chemical Pollution

This is the most widespread form of soil pollution. It includes the accumulation of toxic chemicals such as pesticides, hydrocarbons, industrial solvents, acids, and heavy metals.

3.2 Biological Pollution

Occurs when soil is contaminated with pathogenic microorganisms due to untreated sewage, composted garbage, or animal waste. This can result in disease outbreaks.

3.3 Physical Pollution

This includes the presence of non-biodegradable waste like plastics, glass shards, and synthetic materials. These materials disrupt the soil structure and water absorption capacity.

3.4 Radioactive Pollution

Though less common, radioactive elements from nuclear waste, radiological accidents, or improperly disposed hospital waste can significantly contaminate soil for thousands of years.

4. Impact of Modern Agriculture on Soil

Modern agriculture, though aimed at improving productivity, has adverse effects on the health and sustainability of soil ecosystems.

4.1 Overuse of Chemical Inputs

The excessive application of fertilizers and pesticides leads to toxic accumulation in the soil. This disrupts the natural nutrient cycles and harms beneficial soil organisms such as nitrogen-fixing bacteria.

4.2 Loss of Soil Biodiversity

Monoculture farming practices reduce plant diversity and affect the variety and abundance of microorganisms and insects in the soil. This imbalance weakens the soil's ability to self-repair and regenerate.

4.3 Soil Salinization and Water Logging

Over-irrigation, especially with poor drainage systems, causes the accumulation of salts in the soil, a condition known as salinization, which is harmful to most crops. Waterlogged soils also become compact and anaerobic.

4.4 Soil Erosion and Compaction

Repeated tilling with heavy machinery compacts the soil, reducing its porosity and permeability. Additionally, exposed topsoil is vulnerable to erosion by wind and water.

4.5 Introduction of Genetically Modified Crops (GMOs)

While controversial, some studies suggest that GMOs and associated herbicide use may impact soil microflora, altering the natural balance and nutrient dynamics.

Figure 8: Example of modern agriculture

5. Soil Degradation

Soil degradation refers to the decline in soil quality due to various factors such as pollution, erosion, nutrient loss, and salinity. It compromises the soil's ability to perform essential functions like supporting plant growth, regulating water, and filtering pollutants.

5.1 Types of Soil Degradation

Erosion: The physical removal of topsoil by wind or water. Topsoil is rich in nutrients and organic matter essential for plant growth.

Salinization: Increase in soluble salts due to excessive irrigation, especially in arid regions. Salts make the soil hard and reduce its fertility.

Acidification: Excess use of chemical fertilizers (particularly nitrogen-based) can lower the soil pH, making it acidic and unsuitable for most crops.

Nutrient Depletion: Continuous farming without replenishing lost nutrients leads to soil exhaustion and reduced productivity.

Contamination: Presence of persistent pollutants (pesticides, heavy metals) makes the soil toxic and infertile.

Compaction: Pressing of soil particles due to machinery or overgrazing reduces aeration and water infiltration.

6. Consequences of Soil Pollution and Degradation

Reduced Agricultural Output: Loss of fertility makes the soil less productive, threatening food security.

Contamination of Groundwater: Pollutants from soil can seep into groundwater, affecting drinking water sources.

Biodiversity Loss: Soil pollution kills beneficial microorganisms and insects that are essential for ecological balance.

Increased Natural Disasters: Degraded soils contribute to floods, landslides, and desertification.

Health Hazards: Toxic chemicals can enter the food chain, affecting human and animal health.

Climate Change: Soil stores carbon. Degraded soils release carbon dioxide into the atmosphere, contributing to global warming.

7. Prevention and Control Measures

To mitigate soil pollution and degradation, sustainable practices must be adopted:

Organic Farming: Reducing reliance on chemical fertilizers and pesticides by using compost, green manure, and bio-pesticides.

Proper Waste Management: Safe disposal and recycling of industrial, municipal, and e-waste.

Forestation and Reforestation: Planting trees to bind the soil and prevent erosion.

Crop Rotation and Diversification: Maintaining soil fertility by alternating crops.

Soil Remediation Techniques: Such as bioremediation (using microbes), phytoremediation (using plants), and chemical treatment for pollutant removal.

Public Awareness and Policies: Educating farmers and implementing government regulations to control industrial discharge and promote sustainable land use.

Solid Waste Management

Definition

Solid waste refers to **non-liquid, non-gaseous unwanted materials** produced by human activity. These can be domestic, commercial, biomedical, industrial, or agricultural in origin.

Types of Solid Waste

Type	Examples
Domestic waste	Kitchen garbage, plastic, packaging, bottles
Industrial waste	Ash, chemical residues, scrap metals
Biomedical waste	Syringes, surgical gloves, infected materials
E-waste	Discarded electronics (batteries, monitors, circuit boards)
Agricultural waste	Crop residue, manure, pesticides, husks

Problems Caused by Improper Waste Disposal

- Land and water pollution
- Spread of infectious diseases (due to flies, rats, mosquitoes)
- Soil contamination by heavy metals and toxins
- Greenhouse gas emissions (e.g., CH_4 from landfills)
- Blockage of drainage → urban flooding

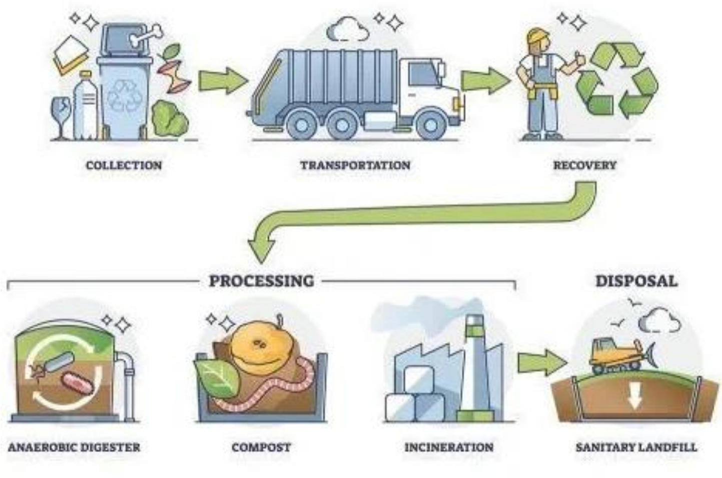

Figure 9: Solid waste management

Solid Waste Management Strategies

1. Waste Minimization at Source

- Avoid plastic bags, use of biodegradable materials

2. Segregation

- Separate biodegradable, recyclable, and hazardous waste at source

3. Treatment and Disposal Techniques

Method	Description
Composting	Organic waste → manure using microbial decomposition
Incineration	Burning waste at high temperature to reduce volume
Sanitary Landfills	Engineered pits with liners to prevent leaching
Recycling	Reprocessing of materials like paper, plastic, and metal
Vermicomposting	Use of earthworms (*Eisenia fetida*) to convert waste to humus

4. Legal Framework

- **Municipal Solid Waste (Management and Handling) Rules, 2000** (India)
- **Swachh Bharat Abhiyan**: Mission to improve cleanliness and waste management

Best Practices

- **3Rs: Reduce, Reuse, Recycle**
- **Public participation** and awareness campaigns
- Use of **smart bins**, IoT for real-time waste tracking (in smart cities)

E-waste and its management

Electronic waste (e-waste) refers to **discarded electrical or electronic devices**, including components, assemblies, and consumables, that are no longer in use or functioning. E-waste is one of the **fastest-growing waste streams** globally due to rapid technological advancements, changing consumer behavior, and shorter product life cycles.

Examples of E-Waste:

- Computers and laptops
- Mobile phones and tablets

- Televisions, monitors
- Refrigerators, washing machines, microwaves
- Batteries, cables, circuit boards, and fluorescent lamps

Figure 10: Various types of E-waste

2. Composition and Character of E-Waste

E-waste is **heterogeneous in nature**, composed of a **complex mix of materials**, some of which are **valuable**, while others are **hazardous**.

2.1 Composition of E-Waste

Material Type	**Common Elements/Components**	**Typical % by weight**
Metals	Iron, copper, aluminum, lead, mercury, cadmium, gold	40–60%
Plastics	PVC, ABS, polycarbonates	20–30%
Glass	CRT glass, fiberglass	5–10%

Material Type	Common Elements/Components	Typical % by weight
Ceramics	Used in resistors, capacitors	2–4%
Hazardous Substances	Lead, mercury, cadmium, brominated flame retardants	Variable
Precious Metals	Gold, silver, palladium, platinum	< 1%

2.2 Valuable Materials in E-Waste

- **Copper**: Found in wires and circuit boards; highly recyclable.
- **Gold and Silver**: Present in connectors, contacts, and chips.
- **Aluminum and Steel**: Found in the casings and frames.
- **Platinum group metals**: Found in hard disks and catalytic converters.

2.3 Hazardous Materials in E-Waste

- **Lead**: Found in CRT monitors, batteries, and solder; causes neurological damage.
- **Mercury**: Used in flat-screen displays and switches; toxic to kidneys and brain.
- **Cadmium**: Present in batteries and chip resistors; carcinogenic.
- **Brominated Flame Retardants (BFRs)**: Found in plastic casings; persistent organic pollutants.
- **Polychlorinated Biphenyls (PCBs)**: Found in capacitors and transformers; toxic and persistent.

3. Characteristics of E-Waste

3.1 Physico-Chemical Characteristics

- High metal content (both **ferrous and non-ferrous**).
- Presence of **toxic heavy metals** and **hazardous chemicals**.
- Non-biodegradable and persistent in the environment.
- High **calorific value** due to plastic content (useful in energy recovery).

3.2 Environmental Characteristics

- E-waste does **not decompose naturally** and may persist for hundreds of years.
- Can **leach hazardous substances** into soil and groundwater if not properly disposed of.
- Improper handling releases **toxic fumes and particles** into the air (e.g., open burning of wires).

4. Environmental and Health Impacts of E-Waste

4.1 Environmental Impacts

- **Soil contamination** due to leaching of heavy metals.
- **Water pollution** when toxins enter water bodies or groundwater systems.
- **Air pollution** through incineration or burning of plastics and wires.

4.2 Health Impacts

- **Neurological damage** from lead and mercury exposure.
- **Respiratory issues** due to inhalation of toxic fumes (e.g., dioxins, furans).
- **Skin disorders, organ damage, and cancer** from exposure to cadmium, chromium, and flame retardants.
- E-waste recycling in **unregulated informal sectors** (especially in developing countries) poses serious risks to workers and nearby residents.

5. E-Waste Management

E-waste management refers to the **collection, recycling, recovery, treatment, and disposal** of e-waste in an environmentally sound manner.

5.1 Steps in E-Waste Management Process

a) Collection and Transportation

- E-waste is collected through formal (authorized dealers) or informal channels.
- Safe and secure transport is required to prevent leakage or damage.

b) Segregation and Dismantling

- Devices are manually dismantled into various components like plastic, metals, circuit boards, etc.

- Reusable parts are separated.

c) Recycling and Recovery

- **Physical recycling**: Shredding and sorting materials (metals, plastics).
- **Chemical recovery**: Leaching or chemical extraction to recover metals like gold and copper.
- **Thermal recovery**: Controlled incineration for energy recovery (requires pollution control systems).

d) Treatment and Disposal

- Non-recyclable hazardous waste is sent to **secured landfills** or **incineration facilities** designed for hazardous waste treatment.

5.2 E-Waste Management Techniques

Method	Description
Recycling	Recovering valuable metals and materials through mechanical or chemical processes.
Reuse and Refurbishing	Extending the life of old devices by repairing and reusing.
Incineration (with energy recovery)	Burning waste in controlled settings to generate energy.
Landfilling (as last resort)	Disposing of residual toxic waste in specially designed landfills.

Global Environmental Problems and Global Efforts

1. Climate Change and Its Impacts on the Human Environment

1.1 What is Climate Change?

Climate change refers to **long-term alterations in temperature, precipitation, wind patterns**, and other elements of the Earth's climate system. The primary cause of modern climate change is **human-induced emissions of greenhouse gases (GHGs)** such as carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O).

1.2 Causes of Climate Change

- **Burning of fossil fuels** (coal, oil, natural gas)
- **Deforestation** (reduces carbon sinks)
- **Industrial emissions**
- **Agricultural practices** (e.g., methane from livestock)
- **Urbanization**

1.3 Impacts on Human Environment

- **Rising temperatures** and heatwaves
- **Melting glaciers** and **rising sea levels**, leading to loss of land and freshwater
- Increased frequency of **extreme weather events**: floods, droughts, cyclones
- **Agricultural disruption**: reduced crop yields, food insecurity
- **Water scarcity**
- Spread of **vector-borne diseases** (e.g., malaria, dengue)
- **Migration and displacement** due to uninhabitable conditions (climate refugees)

2. Ozone Depletion and Ozone-Depleting Substances (ODS)

2.1 What is the Ozone Layer?

The ozone layer is a region in the **stratosphere (10–50 km above Earth)** that contains a high concentration of ozone (O_3). It **absorbs the majority of the sun's harmful ultraviolet (UV-B) radiation**, protecting life on Earth.

2.2 Ozone Depletion

Ozone depletion refers to the **thinning of the ozone layer**, primarily due to man-made chemicals that break down ozone molecules.

2.3 Ozone-Depleting Substances (ODS)

These are compounds that release **chlorine or bromine atoms** when they are broken down by UV radiation in the stratosphere.

ODS	Sources
CFCs (Chlorofluorocarbons)	Refrigerants, air conditioners, aerosol sprays
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl chloroform	Industrial degreasing agent
Methyl bromide	Fumigant in agriculture

2.4 Effects of Ozone Depletion

- Increased UV radiation exposure
- Higher risk of **skin cancer** and **cataracts**
- Damage to **terrestrial and aquatic ecosystems**
- Reduced crop productivity
- Weakening of immune systems in humans and animals

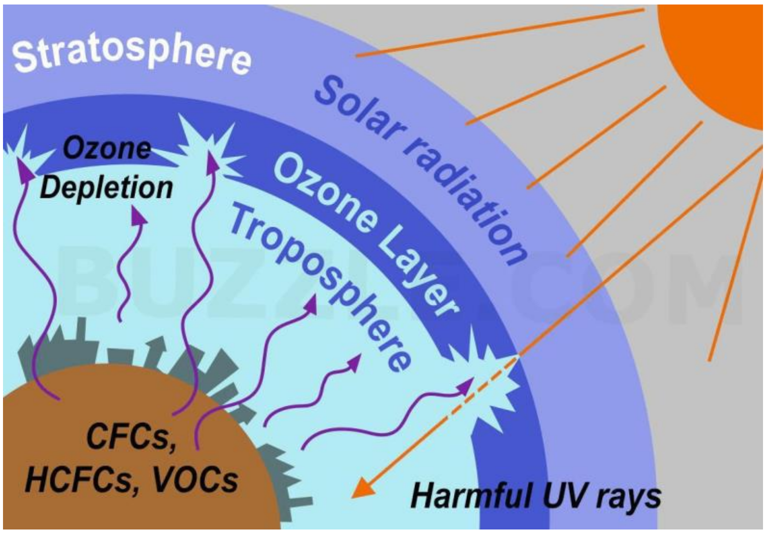

Figure 11: Ozone Depletion and Ozone-Depleting Substances

3. Deforestation and Desertification

3.1 Deforestation

Deforestation is the **removal or clearing of forests** for agriculture, urban development, mining, or logging.

Causes:

- Agriculture expansion (shifting cultivation, plantations)
- Logging for timber and fuel
- Infrastructure development
- Cattle ranching

Impacts:

- Loss of biodiversity
- Soil erosion and reduced fertility
- Disturbance of the water cycle
- Increased GHG emissions (loss of carbon sinks)
- Displacement of indigenous communities

3.2 Desertification

Desertification is the **degradation of land in arid, semi-arid, and dry sub-humid areas** due to climatic variations and human activities.

Causes:

- Overgrazing
- Deforestation
- Unsustainable farming
- Poor irrigation practices
- Climate change (droughts)

Impacts:

- Loss of productive land
- Reduced food and water availability
- Increased poverty and hunger
- Forced migration
- Loss of vegetation cover and biodiversity

4. International Conventions and Protocols

Global environmental issues require **collective international efforts**. Several **treaties and protocols** have been adopted to mitigate these problems.

4.1 Earth Summit (1992) – Rio de Janeiro

Also known as the United Nations Conference on Environment and Development (UNCED).

Key Outcomes:

- Agenda 21: A comprehensive plan of action for sustainable development.
- Rio Declaration: 27 principles guiding sustainable development.
- Formation of UNFCCC (United Nations Framework Convention on Climate Change).
- Convention on Biological Diversity (CBD)
- Forest Principles: Non-legally binding guidelines for forest conservation.

4.2 Kyoto Protocol (1997)

The Kyoto Protocol is an international treaty under the UNFCCC that legally binds developed countries to reduce greenhouse gas emissions.

Key Features:

- Adopted in 1997, entered into force in 2005.
- Set binding targets for 37 industrialized countries.
- Introduced flexible mechanisms:
  - Clean Development Mechanism (CDM)
  - Joint Implementation (JI)
  - Emissions Trading
- Divided countries into:
  - Annex I (industrialized nations with emission targets)
  - Non-Annex I (developing countries without targets)

4.3 Montreal Protocol (1987)

The Montreal Protocol is a global agreement to phase out ozone-depleting substances.

Key Features:

- Signed in 1987; entered into force in 1989.
- Legally binding; ratified by all UN member states.
- Targets elimination of CFCs, halons, carbon tetrachloride, etc.
- Includes amendments to add more substances and strengthen controls.

- The Kigali Amendment (2016) added HFCs (hydrofluorocarbons) to the list.

Success:

- Over 99% of ODS phased out.
- Signs of ozone layer recovery observed.
- A model for successful global environmental cooperation.